

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** (Trimethylsilyl)diazomethane, 2M in hexanes

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>H26744</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Aspiration Toxicity

Category 1

Acute Inhalation Toxicity - Vapors

Category 2

Skin Corrosion/Irritation

Category 2

Carcinogenicity

Category 1B

Reproductive Toxicity

Category 2

Specific target organ toxicity - (single exposure)

Category 3 Category 1

Specific target organ toxicity - (repeated exposure)

Category 2

#### Environmental hazards

Chronic aquatic toxicity

Category 2

### Label Elements



Signal Word

Danger

#### Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H330 - Fatal if inhaled  
H336 - May cause drowsiness or dizziness  
H350 - May cause cancer  
H361 - Suspected of damaging fertility or the unborn child  
H370 - Causes damage to organs  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects

#### Precautionary Statements

##### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection  
P284 - Wear respiratory protection  
P273 - Avoid release to the environment

##### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P308 + P311 - IF exposed or concerned: Call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P391 - Collect spillage

##### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

## Section 3 - Composition and Information on Ingredients

| Component                    | CAS No     | Weight % |
|------------------------------|------------|----------|
| Hexane, branched and linear  | 92112-69-1 | 67       |
| (Trimethylsilyl)diazomethane | 18107-18-1 | 33       |

## **Section 4 - First Aid Measures**

### **Description of first aid measures**

|                                            |                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                   |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                                                                          |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. Risk of serious damage to the lungs (by aspiration). |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                          |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                                                                              |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                                                                      |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                                                                                                                                                                                                                                    |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                              |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Extremely flammable.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Silicon dioxide.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

#### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. To maintain product quality: Keep refrigerated.

#### Incompatible Materials

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

The product does not contain any hazardous materials with occupational exposure limits established.

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

|                        |                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Tight sealing safety goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications) |
| <b>Hand Protection</b> | Protective gloves                                                                                                       |

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>   | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>  | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                  |                          |                                          |
|----------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>            | Liquid                   |                                          |
| <b>Appearance</b>                | Clear Yellow             |                                          |
| <b>Odor</b>                      | mild                     |                                          |
| <b>Odor Threshold</b>            | No data available        |                                          |
| <b>pH</b>                        | No information available |                                          |
| <b>Melting Point/Range</b>       | No data available        |                                          |
| <b>Softening Point</b>           | No data available        |                                          |
| <b>Boiling Point/Range</b>       | 96 °C / 204.8 °F         | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>     | Highly flammable         | On basis of test data                    |
| <b>Flammability (solid,gas)</b>  | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>          | No data available        |                                          |
| <b>Flash Point</b>               | -23 °C / -9.4 °F         | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>  | No data available        |                                          |
| <b>Decomposition Temperature</b> | No data available        |                                          |

|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Viscosity                               | No data available        |             |
| Water Solubility                        | Insoluble                |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Component                               | log Pow                  |             |
| Hexane, branched and linear             | 4.11                     |             |
| Vapor Pressure                          | No data available        |             |
| Density / Specific Gravity              | 0.718                    |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Particle characteristics                | Not applicable (liquid)  |             |

#### Other information

**Explosive Properties** Vapors may form explosive mixtures with air

## Section 10 - Stability and Reactivity

|                                  |                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                                    |
| Stability                        | Stable under normal conditions.                                                                               |
| Sensitivity to Mechanical Impact | No information available                                                                                      |
| Sensitivity to Static Discharge  | No information available                                                                                      |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                      |
| Hazardous Reactions              | None under normal processing.                                                                                 |
| Conditions to Avoid              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition.         |
| Incompatible Materials           | Strong oxidizing agents.                                                                                      |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Nitrogen oxides (NO <sub>x</sub> ). Silicon dioxide. |

## Section 11 - Toxicological Information

#### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| Inhalation | Not an expected route of exposure.                                                        |
| Eyes       | Avoid contact with eyes. May cause irritation.                                            |
| Skin       | Avoid contact with skin. May cause irritation.                                            |
| Ingestion  | May be harmful if swallowed. Harmful if swallowed. Potential for aspiration if swallowed. |

#### Numerical measures of toxicity

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | Based on available data, the classification criteria are not met |
| Inhalation          | Category 2                                                       |

#### Toxicology data for the components

| Component                   | LD50 Oral                  | LD50 Dermal                | LC50 Inhalation                          |
|-----------------------------|----------------------------|----------------------------|------------------------------------------|
| Hexane, branched and linear | LD50 = 15000 mg/kg ( Rat ) | LD50 = 3350 mg/kg (Rabbit) | LC50 = 259354 mg/m <sup>3</sup> (Rat) 4h |

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 1B

(g) reproductive toxicity;  
Reproductive Effects Category 2  
Experiments have shown reproductive toxicity effects on laboratory animals

(h) STOT-single exposure; Category 3  
Results / Target organs Central nervous system (CNS)

(i) STOT-repeated exposure; Category 2  
Target Organs Peripheral Nervous System (PNS), Central nervous system (CNS), Eyes, Respiratory system, Skin, Liver, Blood.

(j) aspiration hazard; Category 1

Other Adverse Effects The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Terrestrial ecotoxicity** There is no data for this product

### **Persistence and Degradability**

**Persistence** Insoluble in water, Persistence is unlikely, based on information available.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component                   | log Pow | Bioconcentration factor (BCF) |
|-----------------------------|---------|-------------------------------|
| Hexane, branched and linear | 4.11    | No data available             |

**Mobility**

Spillage unlikely to penetrate soil. The product is insoluble and floats on water. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. . Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility.

**Other adverse effects**

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## Section 14 - Transport Information

| Component                                        | Hazchem Code |
|--------------------------------------------------|--------------|
| Hexane, branched and linear<br>92112-69-1 ( 67 ) | 3YE          |

**NZS 5433:2020**

UN-No UN1992  
Proper Shipping Name Flammable liquid, toxic, n.o.s.  
Technical Shipping Name (HEXANE, (TRIMETHYLSILYL)DIAZOMETHANE)  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group II

**IATA**

UN-No UN1992  
Proper Shipping Name Flammable liquid, toxic, n.o.s.  
Technical Shipping Name (HEXANE, (TRIMETHYLSILYL)DIAZOMETHANE)  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group II



**IMDG/IMO**

UN-No UN1992  
Proper Shipping Name Flammable liquid, toxic, n.o.s.  
Technical Shipping Name (HEXANE, (TRIMETHYLSILYL)DIAZOMETHANE)  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group II

| Component                                        | IMDG Marine Pollutant                    |
|--------------------------------------------------|------------------------------------------|
| Hexane, branched and linear<br>92112-69-1 ( 67 ) | IMDG regulated marine pollutant (UN1208) |

Environmental hazards Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not applicable, packaged goods

Special Precautions No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

#### International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

| Component                   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                    |
|-----------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------|
| Hexane, branched and linear |                                                                                           |                                                                                          | IMDG regulated marine pollutant (UN1208) |

Authorisation/Restrictions according to EU REACH Not applicable

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                    | CAS No     | NZIoC | AICS | EINECS | ELINCS    | NLP | KECL | IECSC | TCSI |
|------------------------------|------------|-------|------|--------|-----------|-----|------|-------|------|
| Hexane, branched and linear  | 92112-69-1 | X     | X    | -      | 438-390-3 | -   | -    | -     | X    |
| (Trimethylsilyl)diazomethane | 18107-18-1 | -     | -    | -      | -         | -   | -    | -     | X    |

| Component                    | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|------------------------------|------------|------|-----------------------------------------------|-----|------|-------|------|------|
| Hexane, branched and linear  | 92112-69-1 | -    | -                                             | -   | -    | -     | X    | X    |
| (Trimethylsilyl)diazomethane | 18107-18-1 | -    | -                                             | -   | -    | -     | -    | -    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

**Training Advice**

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

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First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 20-Mar-2023  
**Revision Summary** SDS sections updated 1 2 3 7 9

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**